

James R Rose | Resume

Computational Biologist PhD

Status: PhD Candidate, Emory University
Field: Immunology, Genetics, Data Science
Languages: R, bash, Python

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Summary

Computational biologist PhD & former biotech/pharmaceutical bench scientist. Studying gene regulation and epigenetics of the immune system. Looking for opportunities to apply my quantitative & computational analysis skills towards interesting biological questions in translational research.

Research Experience

Graduate Research Assistant - Emory Genetics and Molecular Biology Program

Aug. 2019 - Present

- Integrating gene expression (RNAseq) and epigenetic (ATACseq) data sets from clinical patient groups of a large-scale human study of systemic lupus erythematosus (SLE). Motivating question: Do changes in disease activity affect cellular and molecular phenotypes of B cells in SLE?
- Thesis focus: Integration of multi-omic sequencing datasets in lymphocytes
- Designed and maintained bioinformatic pipelines for next-generation sequencing data processing

Senior Associate Scientist - Janssen Research and Development

Mar. 2016 - Jun. 2019

- Led a team of development researchers on process development, new technology implementation, and regulatory submissions for therapeutic product worth \$6B in annual sales
- Implemented and transferred statistical process control technologies for monoclonal antibody manufacturing at international GMP facilities
- Authored sections of BLA and other regulatory filings

Engineer I - Biogen

Jun. 2013 - Mar. 2016

- Process development of mammalian cell culture processes for monoclonal antibody production at large-scale manufacturing
- Developed advanced process analytics technologies (Raman spectroscopy for remote metabolite monitoring)
- Pioneered use of stochastic (Monte Carlo) simulation modeling for manufacturing facility fit analysis

Education

PhD, Genetics and Molecular Biology - Emory University

Aug. 2019 - Present

- Thesis advisor: Jeremy M Boss. Expected graduation March 2024

AI Innovation Fellowship - Inspirit Innovators

Jul. 2023 - Aug. 2023

- Short course on current techniques in machine learning and AI
- Completed a natural language processing (NLP) project focused on predicting movie genres based on descriptions

B.S. and Masters, Biotechnology - Penn State University

Aug. 2008 - May. 2013

- Integrated masters with B.S. degree
- Thesis: Modeling interfacial shear stress on mammalian cell culture systems

Publications

- Rose, James R; Ishii, Yusho; Hicks, Sakeenah; Scharer, Christopher D; Jenks, Scott A; Sanz, Inaki; Boss, Jeremy M; Clinically inactive SLE resting naïve B cells retain distinct transcriptomes and durable alterations to the epigenome. In preparation NA, NA, 2024
- Rose, James R; AkdoganOzdilek, Bagdeser; Rahmberg, Andrew R; Powell, Michael D; Hicks, Sakeenah L; Scharer, Christopher D; Boss, Jeremy M; Distinct transcriptomic and epigenomic modalities underpin human memory T cell subsets and their activation potential. *Communications Biology* 6, 363, 2023
- Patterson, Dillon G; Kania, Anna K; Price, Madeline J; Rose, James R; Scharer, Christopher D; Boss, Jeremy M; An IRF4MYCmTORC1 integrated pathway controls cell growth and the proliferative capacity of activated B cells during B cell differentiation in vivo. *The Journal of Immunology* 207, 17981811, 2021

Posters

- Rose, James R; Ishii, Yusho; Hicks, Sakeenah; Scharer, Christopher D; Sanz, Ignacio; Boss, Jeremy; Early peripheral B cells in inactive SLE are transcriptionally and epigenetically primed for Tdependent activation. *The Journal of Immunology* 210, 149.08149.08, 2023
- Akdogan, Bagdeser; Rose, James R; Rahmberg, Andrew R; Scharer, Christopher D; Boss, Jeremy M; Human Memory T cells show differential metabolic preferences. *The Journal of Immunology* 210, 87.0587.05, 2023
- Wiggins, Keenan J; Rose, James; Boss, Jeremy M; Scharer, Christopher D; Class switched memory B cell development and transcriptional programming is regulated by EZH2. *The Journal of Immunology* 210, 148.06148.06, 2023
- Powell, Michael Duane; Rose, James R; Guopta, Mansi; Rahmberg, Andrew; Scharer, Christopher D; Boss, Jeremy M; Tissuespecific heterogeneity of Human T follicular helper cells marked by unique transcriptomes and regulatory architecture. *The Journal of Immunology* 210, 148.22148.22, 2023
- Rose, James R; Rahmberg, Andrew R; Powell, Michael D; Scharer, Christopher D; Boss, Jeremy M; Epigenome accessibility changes before and after activation reveal distinct and progressive differentiation for human memory T cell subsets. *The Journal of Immunology* 208, 57.0257.02, 2022
- Powell, Michael Duane; Rose, James R; Neeld, Dennis; Rahmberg, Andrew R; Scharer, Christopher D; Boss, Jeremy M; Defining the cisand transregulatory architecture for human PD1 in T Follicular Helper Cells. *The Journal of Immunology* 208, 112.21112.21, 2022
- Ishii, Yusho; Rose, James; Cashman, Kevin; Hicks, Sakeenah; Deguchi, Tsuneo; Scharer, Christopher; Jenks, Scott; Boss, Jeremy; Sanz, Inaki; Distinct Transcriptomic and Epigenetic Profiles Mediated by Disease ctivity in SLE. *ARTHRITIS & RHEUMATOLOGY* 74, 2223, 2022
- Powell, Michael Duane; Neeld, Dennis Kyle; Rose, James; Scharer, Christopher; Boss, Jeremy M; Exploring Mechanisms of Human PD1 Gene Regulation in T Follicular Helper Cells. *The Journal of Immunology* 206, 53.0653.06, 2021
- Rose, James R; Rahmberg, Andrew; Scharer, Christopher; Boss, Jeremy M; Unique abilities of human memory T cell subsets are associated with both specialized gene expression modules and epigenetic priming.. *The Journal of Immunology* 206, 98.2898.28, 2021

Presentations

- Emory Chromatin Club, “Setting and re-setting chromatin accessibility in inactive SLE”, Sept 14 2023
- American Association of Immunologists, Annual Meeting Washington D.C., “Early peripheral B cells in inactive SLE are transcriptionally and epigenetically primed”, May 12 2013
- American Association of Immunologists, Annual Meeting Portland, Ore., “Epigenome accessibility changes before and after activation reveal distinct and progressive differentiation for human memory T cell subsets”, May 6 2022

Teaching Experience

Course lecturer and TA - IBS519: Stat rigor, reproducibility, and experimental design Fall 2022 and Fall 2023

- Developed teaching materials for revamped course curriculum
- Lecture on data visualization principles and use of ggplot2 in R to incoming graduate students

Teaching assistant - IBS500R: Intro to data analysis using R Fall 2020 and Fall 2021

- Assisted in teaching R programming to incoming graduate students in biomedical graduate programs

Technical Skillset

Programming	Computational Biology	Statistics	Biology
R scripting	Data visualization	Modeling	Immunology
Unix/command line	Bioinformatics	Inference	T cells
Bash programming	RNAseq	Network analysis	B cells
Python	ATACseq		Epigenetics/Gene regulation
Markdown	Git/version control		Autoimmune disease

Awards, Recognition, and Leadership

- Abstract Award, American Association for Immunology Annual Meeting, 2022 and 2023
- Student Representative, Genetics & Molecular Biology Graduate Program
- Eberly College of Science Undergraduate Research Award
- Phi Beta Kappa National Honors Society